

9.ANAESTHESIA WORKSTATION WITH AGM MODULE

TECHNICAL SPECIFICATION

1. GENERAL REQUIREMENTS

Sl. No.	Parameter	Minimum Tender Requirement
1	Equipment	Advanced Anaesthesia Workstation
2	Application	Adult, Paediatric and Neonatal anaesthesia
3	Configuration	Integrated anaesthesia workstation with ventilator, electronic gas mixer and breathing system
4	Ventilator Technology	Electronically driven piston ventilator or equivalent/superior technology
5	Display	Minimum 15-inch colour touchscreen; 15.3-inch preferred
6	Gas Supply	O ₂ , Air and N ₂ O
7	Gas Mixer	Electronic gas mixer with automatic fresh-gas composition control
8	Breathing System	Integrated compact breathing system
9	Fresh Gas Decoupling	Available
10	Breathing Gas Warmer	Integrated
11	Low Flow Anaesthesia	Low-flow and minimal-flow support
12	Vaporizers	Capability for up to 3 vaporizers
13	Self-Test	Automatic/fully automated pre-use system test
14	Patient Categories	Adult, Paediatric and Neonatal
15	Mounting	Mobile trolley configuration; wall/ceiling variants where required
16	Communication	Network connectivity and data communication
17	DICOM/HL7	As applicable for hospital IT integration
18	Safety	Comprehensive electronic and mechanical safety features

2. VENTILATOR

The workstation shall have an electronically controlled piston ventilator or equivalent high-performance ventilator with:

- Precise tidal-volume delivery.

- Active PEEP control.
- High trigger sensitivity.
- Fresh-gas decoupling.
- Integrated compliance compensation.
- Automatic compensation for circuit compliance.
- Suitable ventilation for adult, paediatric and neonatal patients.
- Lung-protective ventilation capability.
- Automatic monitoring of ventilation parameters.

Minimum Ventilation Modes

Mode	Requirement
Manual/Spontaneous	Required
Volume Controlled (VC)	Required
VC-Synchronized + Pressure Support	Required
Pressure Controlled (PC)	Required
PC-Synchronized + Pressure Support	Required
Pressure Support	Required
Volume Controlled AutoFlow	Required
CPAP	Required where available in applicable configuration
APRV	Optional / not mandatory

3. VENTILATION MONITORING

The system shall continuously display and monitor:

- Tidal volume.
- Minute volume.
- Respiratory rate.
- Peak airway pressure.
- Mean airway pressure.
- PEEP.
- Plateau pressure, where applicable.
- FiO₂.
- Compliance.
- Airway resistance, where available.
- Inspiratory and expiratory flow.
- Inspiratory and expiratory volumes.
- End-tidal CO₂ when patient gas measurement is configured.
- Anaesthetic agent concentration, where applicable.

The workstation shall provide **real-time pressure-volume and flow-volume loops**.

4. FRESH GAS SYSTEM

The workstation shall include:

- Electronic gas mixer.
- Oxygen, Air and N₂O gas supply.
- Automatic fresh-gas composition control.
- Fresh-gas flow adjustment.
- Low-flow anaesthesia support.
- Minimal-flow anaesthesia support.
- Low-flow guidance/wizard.
- Visual breathing bag filling-level indication.
- Fresh-gas decoupling.

The official specification lists **O₂, Air and N₂O** as supported gases and an **electronic gas mixer with automatic fresh-gas composition control**, along with low/minimal-flow support.

5. BREATHING SYSTEM

The breathing system shall have:

- Integrated compact breathing system.
- Reusable and/or compatible breathing circuit options.
- Easy access for cleaning and disinfection.
- Tool-free or minimal-tool disassembly.
- CO₂ absorber system.
- Automatic leak detection.
- Breathing system compliance compensation.
- Heated breathing system.
- Water condensation management.
- Manual ventilation capability in case of ventilator failure.
- APL valve for manual ventilation.

6. DISPLAY & USER INTERFACE

The workstation shall have:

- Minimum 15-inch colour touchscreen.
- **15.3-inch touchscreen preferred.**
- High-resolution display.
- Configurable screen layout.
- User-selectable display views.
- Storable screen configurations/profiles.

- Graphical display of waveforms.
- Pressure-time waveform.
- Flow-time waveform.
- Volume-time waveform.
- Pressure-volume loop.
- Flow-volume loop.
- Smart alarm management.
- Visual and audible alarms.
- Alarm priority indication.
- Context-sensitive user interface.

7. PATIENT GAS MONITORING

The workstation should be supplied with an integrated or compatible patient gas measurement module capable of measuring:

- Inspired O₂.
- Expired O₂.
- Inspired CO₂.
- Expired CO₂.
- Inspired anaesthetic agent.
- Expired anaesthetic agent.
- Anaesthetic agent identification.
- MAC value.
- Respiratory rate.

The bidder shall clearly specify whether the **patient gas measurement module is included as standard or offered as an option.**

8. VAPORIZER

The workstation shall be compatible with modern anaesthetic agent vaporizers.

Minimum requirements:

- Provision for at least 2 vaporizers in normal configuration.
- Capability for up to 3 vaporizers, where applicable.
- Interlock system preventing simultaneous activation.
- Secure vaporizer mounting.
- Easy filling system.
- Agent-specific filling system.
- Compatible with commonly used volatile anaesthetic agents.

9. AUTOMATIC SELF-TEST

The workstation shall provide:

- Automatic pre-use self-test.
- Automated leak test.
- Breathing system test.
- Ventilator test.
- Gas supply test.
- Alarm test.
- Sensor test.
- System readiness indication.
- Visual test guidance.
- Test result display.
- Error identification.
- Corrective action guidance.

10. ALARM SYSTEM

The workstation shall provide:

High-Priority Alarms

- Apnoea.
- High airway pressure.
- Low oxygen concentration.
- Low minute volume.
- Disconnection.
- Ventilator failure.
- Gas supply failure.
- Power failure.

Medium/Low-Priority Alarms

- Low battery.
- Low gas pressure.
- CO₂ absorber status.
- Sensor failure.
- Technical fault.
- Maintenance reminder.

The system shall provide:

- Audible alarm.
- Visual alarm.
- Alarm priority classification.
- Alarm history.
- Alarm volume adjustment.

- Alarm limit adjustment.

11. SAFETY FEATURES

The workstation shall incorporate:

- Mechanical/manual ventilation backup.
- O₂ failure protection.
- Hypoxic mixture protection.
- Low gas supply alarm.
- High airway pressure protection.
- Apnoea alarm.
- Disconnect alarm.
- Power failure alarm.
- Battery backup.
- Emergency ventilation capability.
- Automatic system self-test.
- Automatic leak detection.
- APL valve.
- Independent O₂ monitoring.
- Oxygen supply failure protection.

12. BATTERY BACKUP

The workstation shall have:

- Internal rechargeable battery.
- Automatic changeover to battery during mains power failure.
- Battery status indication.
- Low battery alarm.
- Adequate backup for continued ventilation during short-duration power interruptions.

The bidder shall clearly mention:

- Battery type.
- Battery capacity.
- Expected backup duration.
- Battery charging time.
- Battery replacement interval.

13. DATA CONNECTIVITY

The workstation shall have:

- LAN connectivity.
- USB port.
- RS-232/serial communication.
- MEDIBUS.X or equivalent communication protocol.
- Data export capability.
- Network integration.
- Anaesthesia record management compatibility.
- Hospital information system integration.
- Patient monitoring integration.

14. ACCESSORY MANAGEMENT

Where available, the system should support:

- Breathing circuit identification.
- Flow sensor identification.
- Water trap identification.
- CO₂ absorber identification.
- Accessory replacement reminders.
- Incorrect accessory connection alerts.

15. WORKSTATION / TROLLEY

The trolley shall have:

- Four castors/wheels.
- At least two locking brakes.
- Large working surface.
- Lockable drawer.
- Storage shelves.
- Cylinder mounting provision.
- Cable management.
- Integrated electrical outlets.
- Device mounting rails.
- IV pole, where required.
- Adjustable monitor mounting.
- Ergonomic design.
- Easy manoeuvrability.

The bidder shall specify the exact dimensions and weight of the offered configuration.

16. REQUIRED ACCESSORIES

The system shall be supplied as a complete operational unit with:

1. Anaesthesia workstation – **1 No.**
2. Integrated E-Vent/equivalent ventilator – **1 No.**
3. Electronic gas mixer – **1 No.**
4. Integrated breathing system – **1 No.**
5. 15.3-inch touchscreen display – **1 No.**
6. Patient gas module – **1 No.**, preferably included
7. Anaesthetic agent vaporizers – **minimum 2 Nos.**
8. Adult breathing circuit – **2 Nos.**
9. Paediatric breathing circuit – **1 No.**
10. Neonatal breathing circuit – **1 No.**
11. Reusable breathing bag – **2 Nos.**
12. CO₂ absorber – **2 Nos.**
13. Flow sensors – **minimum 2 sets**
14. Water trap – **2 Nos.**
15. Manual resuscitation bag – **1 No.**
16. Breathing hoses and connectors – **1 complete set**
17. O₂ and Air pipeline hoses – **1 set**
18. N₂O pipeline hose – **1 No.**, if N₂O supply is available
19. Power cable – **1 No.**
20. Battery backup – Integrated
21. User manual – **1 No.**
22. Service manual – **1 No.**
23. Installation and commissioning – Included
24. Application training – Included

17. QUALITY & REGULATORY REQUIREMENTS

The bidder shall submit:

- Manufacturer's authorization certificate.
- Original manufacturer's product catalogue.
- Technical datasheet.
- Item-wise point-by-point compliance statement.
- Applicable regulatory approvals.
- Applicable IEC/ISO compliance certificates.
- Electrical safety certificate.
- Calibration certificate.
- Installation qualification documentation.
- User manual.
- Service manual.
- Preventive maintenance schedule.

- Cleaning and disinfection instructions.

The offered equipment shall comply with applicable Indian regulatory requirements and relevant international medical-device standards.

18. WARRANTY

Parameter	Minimum Tender Requirement
Comprehensive Warranty	Minimum 3 years from installation/commissioning
Ventilator	Covered under warranty
Display	Covered under warranty
Gas Mixer	Covered under warranty
Breathing System	Covered under warranty
Patient Gas Module	Covered under warranty
Preventive Maintenance	Minimum 4 visits/year during warranty
Service Response	Within 24–48 hours
Spare Parts	Availability for minimum 7–10 years
Service Support	Authorized service support in India
Training	Anaesthesiologist, OT staff and Biomedical Engineer
Installation	Complete installation and commissioning
Calibration	As per manufacturer's recommended schedule
Software	Updates as applicable during warranty

- **Backup Machine:** In the event of any equipment complaint or breakdown, a suitable backup machine shall be arranged within 12 hours until the original equipment is restored to operational condition.
- **AMC/CAMC Charges:** The AMC/CAMC rate as a percentage of the equipment purchase value, applicable after the warranty period, shall be finalized and frozen at the time of purchase and clearly mentioned in the Purchase Order.
- **Manufacturer Authorization & Warranty:** All authorization letters, warranty documents, and related commitments shall be issued directly by the original manufacturer/company, and not solely by the vendor/dealer.